

SKILL - 1: M. HARSHIKA(2520090047)

1. Create Table

Automatic context help is disabled. Use the tool manually get help for current caret position toggle automatic help

```
1 CREATE DATABASE harshika_2520090047
2 USE harshika_2520090047;
3 CREATE TABLE harshika(
4   roll_no INT PRIMARY KEY,
5   name VARCHAR(50),
6   subject VARCHAR(50),
7   marks DECIMAL(5,2)
8 );
```

#	Time	Action	Message	Duration / Row(s) affected
1	15:37:46	CREATE DATABASE harshika_2520090047	1 row(s) affected	0.094 sec
2	15:40:48	CREATE TABLE harshika(roll_no INT PRIMARY KEY...	Error Code: 1046. No database selected Select the d...	0.015 sec
3	15:41:17	CREATE TABLE harshika(roll_no INT PRIMARY KEY...	Error Code: 1064. You have an error in your SQL synt...	0.000 sec
4	15:41:25	CREATE TABLE harshika(roll_no INT PRIMARY KEY...	Error Code: 1046. No database selected Select the d...	0.000 sec
5	15:42:50	USE harshika_2520090047	0 row(s) affected	0.015 sec
6	15:43:00	CREATE TABLE harshika(roll_no INT PRIMARY KEY...	0 row(s) affected	0.235 sec

2. Insert Data

```
9 INSERT INTO harshika(roll_no, name, subject, marks)
10 VALUES
11 (1, 'Ravi', 'Math', 85.50),
12 (2, 'Sita', 'Math', 92.75),
13 (3, 'Anil', 'Math', 78.40),
14 (4, 'Priya', 'Math', 88.90),
15 (5, 'Vijay', 'Math', 80.25),
16 (6, 'subbusin', 'aws', 98.50),
17 (7, 'dwaraka', 'dbms', 95.75),
18 (8, 'ranjani', 'english', 97.40),
19 (9, 'kavithamam', 'aws1', 99.90),
20 (10, 'seetha', 'azure', 82.25);
```

#	Time	Action	Message
5	12:10:14	CREATE DATABASE harshika_2520090047	1 row(s) affected
6	12:10:32	USE DATABASE harshika_2520090047	Error Code: 1064. You have an error in your SQL s...
7	12:10:45	USE harshika_2520090047	0 row(s) affected
8	12:11:47	CREATE TABLE harshika(roll_no INT PRIMARY KEY, name VARCHAR(50...	0 row(s) affected
9	12:13:17	INSERT INTO harshika(roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Mat...	10 row(s) affected Records: 10 Duplicates: 0 Wa

3. Simple Select

The screenshot shows a SQL IDE interface. At the top, a query editor contains the SQL statement: `SELECT * FROM harshika;`. Below the editor, a toolbar includes options for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. The 'Result Grid' is active, displaying a table with the following data:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25
NULL	NULL	NULL	NULL

Below the result grid, a tab labeled 'harshika 1' is visible. The 'Output' pane shows the 'Action Output' table, which lists the execution of the query and its results:

#	Time	Action	Message
5	12:10:14	CREATE DATABASE harshika_2520090047	1 row(s) returned
6	12:10:32	USE DATABASE harshika_2520090047	Error (0)
7	12:10:45	USE harshika_2520090047	0 row(s) returned
8	12:11:47	CREATE TABLE harshika(roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), marks DECIMAL(5,2))	0 row(s) returned
9	12:13:17	INSERT INTO harshika(roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.75), (3, 'Anil', 'Math', 78.40), (4, 'Priya', 'Math', 88.90), (5, 'Vijay', 'Math', 80.25), (6, 'subbusir', 'aws', 98.50), (7, 'dwaraka', 'dbms', 95.75), (8, 'ranjani', 'english', 97.40), (9, 'kavithamam', 'aws1', 99.90), (10, 'seetha', 'azure', 82.25)	10 row(s) returned
10	12:15:16	SELECT * FROM harshika LIMIT 0, 1000	10 row(s) returned

4. Aggregate Functions

The screenshot shows a SQL IDE interface. The query editor contains several SQL statements using aggregate functions:

```
22 • SELECT * FROM harshika;
23
24 • SELECT COUNT(*) AS total_students FROM harshika;
25 • SELECT SUM(marks) FROM harshika;
26 • SELECT AVG(marks) FROM harshika;
27 • SELECT MAX(marks) FROM harshika;
28 • SELECT MIN(marks) FROM harshika;
```

The 'Result Grid' is active, displaying the results of the aggregate functions:

MIN(marks)
78.40

Below the result grid, a tab labeled 'Result 7' is visible. The 'Output' pane shows the 'Action Output' table, which lists the execution of the aggregate functions and their results:

#	Time	Action	Message
12	12:17:04	SELECT SUM(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned
13	12:17:54	SELECT AVG(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned
14	12:18:46	SELECT MAX(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned
15	12:18:51	SELECT MIN(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned
16	12:19:02	SELECT MAX(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned
17	12:19:02	SELECT MIN(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned

5. Filtering with WHERE

```
30 • SELECT * FROM harshika WHERE marks > 85;
31 • SELECT * FROM harshika WHERE marks >= 90;
32 • SELECT * FROM harshika WHERE marks < 80;
33 • SELECT * FROM harshika WHERE marks BETWEEN 80 AND 90;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wr

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
5	Vijay	Math	80.25
10	seetha	azure	82.25
...	...	...	...

harshika 10 harshika 11 harshika 12 harshika 13 ×

Output

Action Output

#	Time	Action	Message
17	12:19:02	SELECT MIN(marks) FROM harshika LIMIT 0, 1000	1 row(s) returned
18	12:20:29	SELECT * FROM harshika WHERE marks > 85 LIMIT 0, 1000	7 row(s) returned
19	12:20:50	SELECT * FROM harshika WHERE marks > 85 LIMIT 0, 1000	7 row(s) returned
20	12:20:50	SELECT * FROM harshika WHERE marks >= 90 LIMIT 0, 1000	5 row(s) returned
21	12:20:50	SELECT * FROM harshika WHERE marks < 80 LIMIT 0, 1000	1 row(s) returned
22	12:20:50	SELECT * FROM harshika WHERE marks BETWEEN 80 AND 90 LIMIT ...	4 row(s) returned

```
33 • SELECT * FROM harshika WHERE marks BETWEEN 80 AND 90;
34 • SELECT * FROM harshika WHERE name LIKE 'P%';
35 • SELECT * FROM harshika WHERE name IN ('Ravi', 'Sita', 'Vijay');
36 • SELECT * FROM harshika WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
...	...	...	...

harshika 14 harshika 15 harshika 16 × Apply

Output

Action Output

#	Time	Action	Message
20	12:20:50	SELECT * FROM harshika WHERE marks >= 90 LIMIT 0, 1000	5 row(s) returned
21	12:20:50	SELECT * FROM harshika WHERE marks < 80 LIMIT 0, 1000	1 row(s) returned
22	12:20:50	SELECT * FROM harshika WHERE marks BETWEEN 80 AND 90 LIMIT ...	4 row(s) returned
23	12:21:17	SELECT * FROM harshika WHERE name LIKE 'P%' LIMIT 0, 1000	1 row(s) returned
24	12:21:17	SELECT * FROM harshika WHERE name IN ('Ravi', 'Sita', 'Vijay') LIMIT 0...	3 row(s) returned
25	12:21:17	SELECT * FROM harshika WHERE marks > 85 AND (subject = 'Math' OR...	3 row(s) returned

6. Update Queries

```
38 • UPDATE harshika SET marks = 90.00 WHERE roll_no = 3;
39 • UPDATE harshika SET subject = 'Remedial Math' WHERE marks < 80;
40
41 • SET SQL_SAFE_UPDATES = 0;
42 • UPDATE harshika
43   SET subject = 'Remedial Math'
44   WHERE marks < 80;
```

Output

Action Output

#	Time	Action	Message
✗ 27	12:22:38	UPDATE harshika SET subject = 'Remedial Math' WHERE marks < 80	Error Code: 1175. You are using safe update mode and you tried
✓ 28	12:29:21	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
✗ 29	12:29:39	UPDATE student SET marks = 90 WHERE name = 'Harshika'	Error Code: 1146. Table 'harshika_2520090047.student' doesn't
✓ 30	12:30:40	UPDATE harshika SET subject = 'Remedial Math' WHERE marks < 80	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 31	12:30:47	UPDATE harshika SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
✓ 32	12:30:47	UPDATE harshika SET subject = 'Remedial Math' WHERE marks < 80	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0

7. Delete Queries

```
45
46 • DELETE FROM harshika WHERE roll_no = 5;
47 • DELETE FROM harshika WHERE marks < 75;
```

Output

Action Output

#	Time	Action	Message
✗ 29	12:29:39	UPDATE student SET marks = 90 WHERE name = 'Harshika'	Error Code: 1146. Table 'harshika_2520090047.s
✓ 30	12:30:40	UPDATE harshika SET subject = 'Remedial Math' WHERE marks < 80	0 row(s) affected Rows matched: 0 Changed: 0
✓ 31	12:30:47	UPDATE harshika SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0
✓ 32	12:30:47	UPDATE harshika SET subject = 'Remedial Math' WHERE marks < 80	0 row(s) affected Rows matched: 0 Changed: 0
✓ 33	13:32:49	DELETE FROM harshika WHERE roll_no = 5	1 row(s) affected
✓ 34	13:32:55	DELETE FROM harshika WHERE marks < 75	0 row(s) affected

8. Sorting

```
49 • SELECT * FROM harshika ORDER BY marks ASC;
50 • SELECT * FROM harshika ORDER BY marks DESC;
51 • SELECT * FROM harshika ORDER BY name ASC;
52 • SELECT * FROM harshika ORDER BY name DESC;
53
```

Result Grid

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00
*	NULL	NULL	NULL

harshika 17 harshika 18 harshika 19 harshika 20 x

Output

Action Output

#	Time	Action	Message
33	13:32:49	DELETE FROM harshika WHERE roll_no = 5	1 row(s) affected
34	13:32:55	DELETE FROM harshika WHERE marks < 75	0 row(s) affected
35	13:34:26	SELECT * FROM harshika ORDER BY marks ASC LIMIT 0, 1000	9 row(s) returned
36	13:34:26	SELECT * FROM harshika ORDER BY marks DESC LIMIT 0, 1000	9 row(s) returned
37	13:34:26	SELECT * FROM harshika ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
38	13:34:26	SELECT * FROM harshika ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned

9. Group By

```
54 • SELECT subject, SUM(marks) AS total_marks FROM harshika GROUP BY subject;
55 • SELECT subject, AVG(marks) AS avg_marks FROM harshika GROUP BY subject;
56 • SELECT subject, COUNT(*) AS num_students FROM harshika GROUP BY subject;
57
```

Result Grid

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

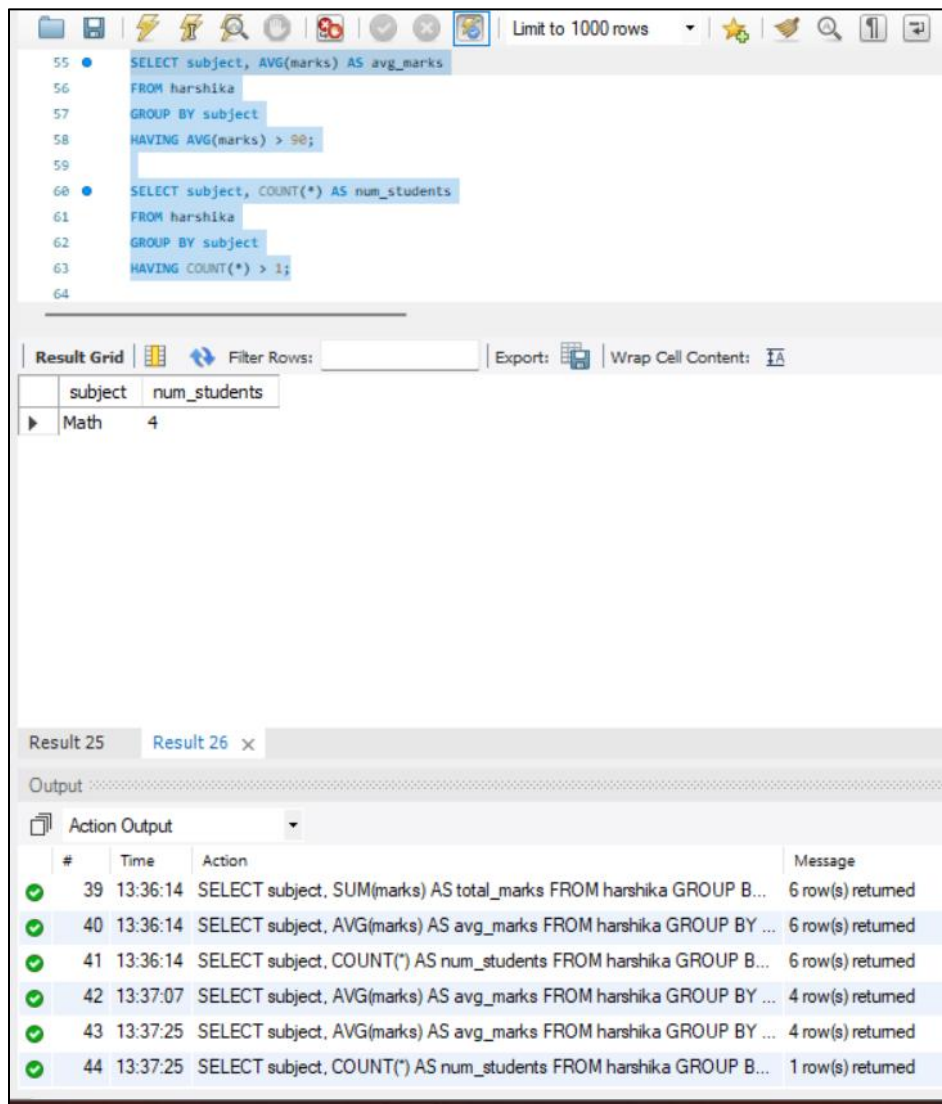
Result 21 Result 22 Result 23 x

Output

Action Output

#	Time	Action	Message
36	13:34:26	SELECT * FROM harshika ORDER BY marks DESC LIMIT 0, 1000	9 row(s) returned
37	13:34:26	SELECT * FROM harshika ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
38	13:34:26	SELECT * FROM harshika ORDER BY name DESC LIMIT 0, 1000	9 row(s) returned
39	13:36:14	SELECT subject, SUM(marks) AS total_marks FROM harshika GROUP BY subject	6 row(s) returned
40	13:36:14	SELECT subject, AVG(marks) AS avg_marks FROM harshika GROUP BY subject	6 row(s) returned
41	13:36:14	SELECT subject, COUNT(*) AS num_students FROM harshika GROUP BY subject	6 row(s) returned

10. Having`



The screenshot displays a SQL IDE interface with two queries and their results.

Query 1 (Line 55-58):

```
SELECT subject, AVG(marks) AS avg_marks
FROM harshika
GROUP BY subject
HAVING AVG(marks) > 90;
```

Query 2 (Line 60-63):

```
SELECT subject, COUNT(*) AS num_students
FROM harshika
GROUP BY subject
HAVING COUNT(*) > 1;
```

Result Grid:

	subject	num_students
▶	Math	4

Output Pane (Action Output):

#	Time	Action	Message
✓ 39	13:36:14	SELECT subject, SUM(marks) AS total_marks FROM harshika GROUP B...	6 row(s) returned
✓ 40	13:36:14	SELECT subject, AVG(marks) AS avg_marks FROM harshika GROUP BY ...	6 row(s) returned
✓ 41	13:36:14	SELECT subject, COUNT(*) AS num_students FROM harshika GROUP B...	6 row(s) returned
✓ 42	13:37:07	SELECT subject, AVG(marks) AS avg_marks FROM harshika GROUP BY ...	4 row(s) returned
✓ 43	13:37:25	SELECT subject, AVG(marks) AS avg_marks FROM harshika GROUP BY ...	4 row(s) returned
✓ 44	13:37:25	SELECT subject, COUNT(*) AS num_students FROM harshika GROUP B...	1 row(s) returned